

3.4.1 Data Types in Python

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 08 of 24

★ Golden

3.4.1 Data Types in Python ★ Golden

LEARN IT

A **data type** specifies the **kind of data** a variable holds — a whole number, a decimal, text, True/False, or a collection of items. The type decides what you can do with the data (you can divide numbers, but not names). Python has several built-in data types:

Name	Type	Description	Example
Integer	<code>int</code>	Whole numbers without decimals (positive, negative or zero).	<code>age = 16</code>
Floating point	<code>float</code>	Numbers with a decimal point.	<code>height = 5.7</code>
String	<code>str</code>	Text data, enclosed in quotes (" " or ' ').	<code>name = "Sammy" or year = "2026"</code>
Boolean	<code>bool</code>	Logical value: True or False (capital T / F).	<code>is_student = True</code>
List	<code>list</code>	Ordered, changeable collection in square brackets [].	<code>marks = [85, 90, 75]</code>
Tuple	<code>tuple</code>	Ordered, unchangeable sequence in parentheses ().	<code>coords = (10, 20)</code>
Dictionary	<code>dict</code>	Key : value pairs in curly braces { }.	<code>student = {"name": "Ahmed", "age": 16}</code>
Set	<code>set</code>	Unordered collection of unique items (no duplicates).	<code>vowels = {'a', 'e', 'i'}</code>

Note: the slide abbreviates tuple as "tup"; the real Python type name is `tuple`, as the program below shows. The slides describe dictionaries as unordered; since Python 3.7 a dictionary remembers the order in which keys were added, but you always look items up **by key**, not by position.

WORKED EXAMPLE

Checking every type with `type()` :

```
age = 16
height = 5.7
name = "Sammy"
year = "2026"
is_student = True
marks = [85, 90, 75]
coords = (10, 20)
student = {"name": "Ahmed", "age": 16}
vowels = {'a', 'e', 'i'}

print(type(age))
print(type(height))
print(type(name), type(year))
print(type(is_student))
print(type(marks))
print(type(coords))
print(type(student))
print(type(vowels))
```

OUTPUT

```
<class 'int'>
<class 'float'>
<class 'str'> <class 'str'>
<class 'bool'>
<class 'list'>
<class 'tuple'>
<class 'dict'>
<class 'set'>
```

Notice that `year = "2026"` is a **string**, not a number, because it is inside quotes.

Working with the collection types:

```
marks = [85, 90, 75]
marks[0] = 88          # lists are changeable
marks.append(60)
print(marks)

coords = (10, 20)
print(coords[0])        # read item at position 0

student = {"name": "Ahmed", "age": 16}
print(student["name"])  # look up by key

nums = {1, 2, 2, 3, 3, 3} # duplicates are removed
print(nums)
print(len(nums))
```

OUTPUT

```
[88, 90, 75, 60]
10
Ahmed
{1, 2, 3}
3
```

Positions (called **indexes**) start at **0**, so `coords[0]` is the first item. Trying to change a tuple causes an error:

```
coords = (10, 20)
coords[0] = 50
```

OUTPUT

```
Traceback (most recent call last):
  File "tuple_test.py", line 2, in <module>
    coords[0] = 50
    ~~~~~^
TypeError: 'tuple' object does not support item assignment
```

REMEMBER IT

Brackets decide the collection: `[ ]` List (changeable), `( )` Tuple (locked), `{ }` with **key:value** = Dictionary, `{ }` with single items = Set (unique). Think: a **list** is a shopping list you can edit; a **tuple** is your date of birth — it never changes.

COMMON MISTAKE

`True` and `False` must start with a capital letter. `true` is not a Boolean; Python treats it as an unknown variable name and gives a `NameError`. Also, `"16"` (string) and `16` (int) are different: `"16" + "1"` gives `"161"`, while `16 + 1` gives `17`.

اردو میں سمجھیں

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔ `int` مکمل اعداد، `float` اعشاریہ والے اعداد، `str` متن، اور `bool` صرف `True` یا `False` کے لیے ہے۔ لسٹ (List) تبدیل ہو سکتی ہے، ٹپل (Tuple) تبدیل نہیں ہو سکتا، ڈکشنری (Dictionary) میں کی اور ویلیو کے جوڑے ہوتے ہیں، اور سیٹ (Set) میں کوئی چیز دہرائی نہیں جاتی۔

EXAM TIP

Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — `[ ]`, changeable, e.g. `marks = [85, 90]`; tuple — `( )`, unchangeable, e.g. `coords = (10, 20)`.

CHECK YOURSELF

1. What is the data type of each? `3.0`, `"3"`, `3`, `False`, `[3]`
2. Predict the output: `print(len({5, 5, 6, 6, 6}))`
3. Which type should store a student's CNIC/B-Form number such as `"42101-1234567-1"`? Why?

Answers: 1. float, str, int, bool, list. 2. 2 (the set keeps only 5 and 6). 3. str, because it contains hyphens and you never do arithmetic on it.

Next lecture: 3.4.2 Input and Output Handling